

TEST-06-INFRASTRUCTURE-TESTING

TEST-06: INFRASTRUCTURE TESTING PROTOCOL

Carrington Storm Motors — Project AEGIS

Bridge Bearings • Substation Transformers • Pipeline Isolation • Railway Signals • Power Line Monopoles

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Reference Standards: AASHTO LRFD, IEEE 693, NACE SP0169, AREMA, NERC TPL-007

INFRASTRUCTURE TESTING PHILOSOPHY

Infrastructure testing differs fundamentally from product testing: you cannot put a substation in a TEM cell, and you cannot cycle a bridge bearing through 100 thermal cycles on demand. Infrastructure validation therefore relies on a combination of laboratory-scale component testing, in-situ pre/post measurement, and long-term continuous monitoring. The protocol defined here aligns with the specific regulatory and operational requirements of each infrastructure asset class.

PART 1: BRIDGE BEARING DIELECTRIC PADS (AASHTO Compliant)

1.1 Product Description

Replacement bridge bearing pads (elastomeric or pot bearing type) with integrated Si₃N₄/PTFE dielectric layer achieving >50 kV/mm dielectric strength, 500-tonne load capacity, and >10⁸ Ω electrical isolation between bridge deck and substructure.

1.2 Laboratory Testing

Test	Standard	Specimen	Acceptance
Dielectric Strength	ASTM D149 (short-time)	150×150 mm bearing pad, full thickness (25–50 mm)	>50 kV/mm (through-thickness); no surface flashover
Compressive Stiffness	AASHTO LRFD 14.7.5 (elastomeric bearings)	Full-scale bearing assembly	Stiffness within ±15% of design; creep <5% after 1,000 hours at 1.5× design load
Shear Modulus	AASHTO LRFD 14.7.5	2 bearings tested in shear	Shear modulus within ±15% of design
Fatigue	2 million cycles at ±50% design shear strain, 3× design compressive load	2 bearings	No cracking; dielectric layer intact (post-test resistance >10 ⁸ Ω)
Low-Temperature	–35°C per AASHTO LRFD Zone requirement	5 bearings	No stiffening beyond design limit; dielectric layer intact
Environmental Durability	Ozone (ASTM D1149, 100 pphm, 100h); UV (ASTM G154, 2,000h); salt spray (ASTM B117, 500h)	5 bearings per exposure	No surface cracking; dielectric resistance >10 ⁷ Ω post-exposure

1.3 In-Situ Pre/Post Measurement (Pilot Bridges)

Measurement	Method	Before Installation	After Installation	Target Change
GIC Current Through Bearing	Rogowski coil around bearing assembly	Current through existing bearing during geomagnetic storm (baseline)	Current through dielectric bearing during equivalent storm	>90% reduction
Bearing Resistance	Megohmmeter (1,000 V) across bearing	Baseline (typically <1 Ω for steel bearing in contact with deck/rebar)	Post-installation resistance deck-to-pier	>10 ⁸ Ω
Structural Strain	FBG strain gauges on bridge deck and pier	Baseline structural response	Post-installation structural response	No change (bearing maintains structural function while adding dielectric isolation)

PART 2: SUBSTATION TRANSFORMER HARDENING

2.1 Product Description

Transformer neutral blocking system using series capacitor or active compensation to prevent GIC from entering transformer windings through the neutral-ground connection.

2.2 Testing Protocol

Test	Method	Acceptance
Neutral Current Blocking	Inject 0.001–1 Hz AC current (1–100 A) into transformer neutral in controlled test; measure winding current with and without blocking device	Neutral current reduction >90%; winding current reduced below saturation threshold
Capacitor Bank Endurance	10,000 cycles of GIC-simulated neutral current at 50 A RMS	Capacitance change <5%; no dielectric breakdown; thermal rise <30°C above ambient
Harmonic Analysis	FFT of transformer primary current during GIC injection — GIC causes half-cycle saturation → harmonics (especially 2nd and 4th) → measure reduction with blocking device	Harmonic current reduction >80% (2nd harmonic <5% of fundamental)
Reactive Power Consumption	Measure transformer VAR consumption with and without blocking device during GIC injection	VAR increase <20% of baseline (unprotected transformer can draw 2–4× rated VAR during GIC half-cycle saturation)
Fault Tolerance	Simulate line-to-ground fault on transmission line while blocking device is active — verify blocking device does not interfere with protection relay operation (fault must be cleared within normal time)	Fault cleared within protection relay time setting (typically 3–6 cycles / 50–100 ms); blocking device auto-bypasses on overcurrent
Lightning Impulse	1.2/50 μ s impulse, 550 kV BIL (Basic Insulation Level) for 230 kV transformer	No damage to blocking device; device re-engages automatically after impulse
Temperature Range	–40°C to +55°C ambient; blocking device typically installed in substation control house (temperature-controlled) or outdoor enclosure	Full function across temperature range; capacitor value within $\pm 10\%$ of nominal at temperature extremes

Cost per Substation: \$380K (equipment + installation + testing); Duration: 6 months

PART 3: PIPELINE ISOLATION SYSTEM (GFRP Dielectric Joints)

3.1 Product Description

GFRP dielectric isolation joints installed at 500 m intervals along buried steel pipeline to break continuous conductive paths that would otherwise carry GIC. Joints must maintain pipeline pressure integrity, allow cathodic protection current to pass through the soil (around the joint, not through it), and withstand ground movement.

3.2 Testing Protocol

Test	Method	Acceptance
Hydrostatic Pressure	1.5× Maximum Allowable Operating Pressure (MAOP) for 24 hours	Zero pressure drop; zero leakage; no structural deformation
Dielectric Isolation	Megohmmeter at 1,000 V across joint (pipeline segment A to segment B through joint)	Resistance >10 ⁷ Ω (lower than building products because pipeline environment is wet/dirty — 10 ⁷ Ω is sufficient to block GIC while allowing static dissipation)
Cathodic Protection (CP) Compatibility	Measure pipe-to-soil potential on both sides of dielectric joint with CP system operational	Potential on both sides within NACE SP0169 criterion (−850 mV vs. Cu/CuSO ₄ , polarized); CP current must not bridge the dielectric joint — current path is through soil, not through joint
Ground Movement	Apply 50 mm differential settlement across joint (simulating soil movement)	No leakage; resistance stays >10 ⁷ Ω; joint flexibility accommodates movement
Thermal Expansion	Cycle joint from −20°C to +60°C (buried pipeline temperature range)	No leakage; seal integrity maintained
Bending Moment	Apply bending moment equivalent to 1,000 m pipeline span (soil settlement)	No structural failure; joint maintains pressure and dielectric isolation

PART 4: RAILWAY SIGNAL PROTECTION (PMMA POF Signal Cables)

4.1 Product Description

Replacement of copper track circuit signal cables with PMMA POF to eliminate GIC induction in railway signaling systems, which can cause false “track occupied” or “track clear” signals — a potentially catastrophic failure mode.

4.2 Testing Protocol

Test	Method	Acceptance
Data Transmission	BER test at 1 Gbps over 1 km POF with 10 connectors (simulated trackside installation)	BER <10 ^{−12} ; latency <100 μs/km
Track Circuit Integrity	Install POF-based track circuit on test track; verify train detection with locomotive at various speeds (0–125 mph)	Train detected within 1 second of entering block; no false clears; no false occupancies
GIC Immunity	Inject 0.001–1 Hz current into rails (simulating GIC) while POF track circuit is active	Zero data errors; zero false occupancy signals (copper-based track circuits can false-occupy at >5 A GIC in rails)
	10 kA, 8/20 μs surge applied to rails — verify POF system survives (POF	Transceivers survive without

Lightning Surge	is inherently immune to lightning — this tests the optoelectronic transceivers at endpoints)	damage; system returns to normal operation within 1 second
Vandalism Resistance	Cut POF cable; verify system reports cable break immediately and defaults to “occupied” (fail-safe)	Alarm within 1 second; block defaults to occupied
Temperature Range	–40°C to +70°C (trackside enclosure temperature)	BER within spec at all temperatures
AREMA Compliance	All tests per AREMA C&S Manual Part 11.5.1 (signal cable requirements)	Pass all AREMA criteria

PART 5: POWER LINE MONOPOLE (BFRP Tower)

5.1 Product Description

BFRP composite monopole transmission tower (30–60 m height) with ceramic insulator strings and MXene-coated conductor attachment points — eliminates the steel tower as a GIC pathway from shield wire to ground.

5.2 Testing Protocol

Test	Method	Acceptance
Cantilever Load	Apply horizontal load at conductor attachment points equivalent to NESC Heavy loading district (ice + wind)	Deflection within allowable; no structural damage; no cracking
Dielectric Isolation	Measure resistance from each conductor attachment point to ground	$>10^8 \Omega$ (standard steel tower: $\sim 10 \Omega$ through ground grid)
Full-Scale Structural Test	Destructive test to failure at full-scale test facility (e.g., EPRI Transmission Line Research Facility)	Failure load $>1.5\times$ NESC extreme wind load; failure mode is progressive (fiber breakage), not catastrophic (brittle fracture)
Lightning Attachment	Apply 100 kA, 8/20 μ s impulse to shield wire — verify lightning current path through dedicated down conductor (MXene-coated BFRP strip), not through structural BFRP (BFRP is insulating at power frequency but can carbon-track at lightning currents — the dedicated down conductor is designed for lightning)	No structural damage; down conductor intact; resistance down conductor-to-ground $<10 \Omega$
UV Resistance	ASTM G154, 5,000 hours (extended — outdoor structure, 25+ year life)	Tensile strength retained $>80\%$; no surface crazing or fiber blooming; YInMn Blue pigment colorfastness $\Delta E < 5$
Vibration (Aeolian)	Simulate wind-induced vibration at cable natural frequency (typically 5–50 Hz) for 10^7 cycles	No fatigue cracking; no loosening of bolted connections
Foundation Integration	Full-scale foundation pull-out test with geopolymers concrete anchor block	Anchor block capacity $>1.5\times$ design uplift; geopolymer-concrete bond >5 MPa

PART 6: IN-SITU GIC MEASUREMENT PROTOCOL

6.1 Before/After Methodology for All Infrastructure Pilots

For each infrastructure pilot site (substation, bridge, pipeline, railway):

1. **Pre-Installation Baseline (3 months):** Install GIC sensors at pilot sites for 3 months before any CSM hardening is installed. Record GIC during geomagnetic activity.
2. **Installation:** Install CSM hardening products with sensors in place.
3. **Post-Installation Monitoring (12 months):** Continue recording GIC. Compare post- vs. pre-installation during storms of equivalent magnitude (matched by NOAA Kp or Dst index).
4. **GIC Reduction Calculation:**
 - Match storm events pre/post by Kp index (± 0.5) or Dst (± 20 nT)
 - For each matched pair, calculate: $\text{Reduction} = (\text{GIC}_{\text{pre}} - \text{GIC}_{\text{post}}) / \text{GIC}_{\text{pre}} \times 100\%$
 - Report mean reduction with 95% confidence interval

6.2 Long-Term Monitoring (5+ Years)

After the 12-month pilot, infrastructure sites transition to ongoing monitoring: - Sensors remain installed permanently - Data feeds into CSM Digital Twin (LandOLil v10+) - Annual reports published: GIC events, product performance, maintenance required - Data shared with utility partners, NERC, and IEEE working groups

INFRASTRUCTURE TESTING BUDGET SUMMARY

Product	Testing Cost	Timeline	Status
Bridge Bearing Pads	\$680K (lab + 3 bridge in-situ)	18 months	◦ Design phase
Substation Transformer Hardening (5 sites)	\$1.9M (\$380K/site)	24 months	◦ Partner identification
Pipeline Isolation (2 sections, 10 km each)	\$1.2M	18 months	◦ Partner identification
Railway Signal Protection (1 test segment)	\$850K	18 months	◦ Partner identification
Power Line Monopole (full-scale test)	\$1.4M (EPRI facility + test pole)	24 months	◦ Design phase
In-Situ Monitoring Equipment (all sites)	\$2.5M	12+ months	◦ Procurement
TOTAL	\$8.53M		

“Infrastructure is the skeleton of civilization. Protecting it from GIC is not about a single product — it is about a systematic approach to identifying, measuring, and eliminating every conductive path that can carry induced current from the Earth into our bridges, substations, pipelines, and railway signals. The testing protocol defined here is the diagnostic tool for that systematic approach. Before we harden, we measure. After we harden, we measure again. The difference is the proof.” — Engineer Orion Cross (SILENCE)

Document generated 2026-07-15. All infrastructure testing is contingent on securing utility and government agency partners (see ROAD-09). In-situ GIC monitoring equipment procurement to begin Q1 2027. Full-scale monopole testing at EPRI facility requires 12-month advance scheduling due to high demand — target test date Q2 2028.